# Project Planning & Management

### End-to-End Business Analytics Decision Platform – Tourism & Hospitality Initiative

### DEPI – Digital Egypt Pioneers Initiative | Track: Data Science

### Trainer: Sara Ashraf

Repository: https://github.com/marioma-123/business-analytics-decision-platform

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## 1. Project Proposal

### 1.1 Overview

This project builds an end-to-end analytics and decision platform for a tourism and hospitality business that manages short-term rental accommodation across several locations. It integrates data from Inside Airbnb, World Bank World Development Indicators, Wikipedia, and the Open-Meteo historical weather API; analyses drivers of prices, demand indicators, and guest satisfaction; predicts nightly listing prices with machine learning; and delivers results through an interactive dashboard and a deployed prediction service.

### 1.2 Problem Statement

A company has data scattered across multiple sources but lacks an integrated analytical solution for monitoring performance, understanding its drivers, and predicting future outcomes. Short-term rental listing records, historical weather, country-level economic indicators, and web-collected landmark/location information are stored separately. Management therefore cannot easily:

- Monitor prices, availability, and ratings across locations and over time.
- Understand drivers of price and demand indicators, including location, property/room type, season, weather, and guest reviews.
- Predict nightly price per listing to support pricing and planning decisions.

### 1.3 Project Objective

Build a complete data science solution following the CRISP-DM lifecycle, from business understanding through deployment and communication.

- Define the business problem, stakeholders, KPIs, analytical questions, and success criteria.
- Collect data from a SQL database (Inside Airbnb files), REST APIs (Open-Meteo and World Bank indicators), and Wikipedia web scraping.
- Integrate and clean data into an analysis-ready dataset.
- Explore descriptive statistics, distributions, correlations, and grouped analyses.
- Build an interactive dashboard with executive KPIs, trends, comparisons, and geographic visualization.
- Train and compare Linear/Ridge Regression, Random Forest, and Gradient Boosting models to predict nightly listing prices.
- Evaluate models and justify the final selection and limitations.
- Track experiments with MLflow, serve predictions with FastAPI, package with Docker, automate checks with GitHub Actions, and deploy to a cloud platform (provider undecided).
- Communicate findings through an executive summary, technical report, dashboard, and stakeholder presentation.

## 1.4 Scope

| CRISP-DM phase | Deliverable |
|---|---|
| 1. Business Understanding | Business problem, stakeholders, KPIs, analytical questions, success criteria |
| 2. Data Collection | SQL database (Inside Airbnb), REST APIs (Open-Meteo and World Bank), Wikipedia scraping |
| 3. Data Preparation | Integration, missing values, formatting, normalization, categorical transformation |
| 4. EDA | Descriptive statistics, distributions, correlations, group-by analysis |
| 5. Visualization | Executive KPIs, trend and comparative charts, map, interactive dashboard |
| 6. Machine Learning | Multiple regression models for nightly price prediction |
| 7. Evaluation | Metrics, model comparison, limitations, selection rationale |
| 8. Deployment | MLflow, FastAPI, Docker, GitHub Actions, cloud hosting (provider undecided) |
| 9. Communication | Executive summary, technical report, dashboard, stakeholder presentation |

## 1.5 Stakeholder

| Stakeholder | Role | Main interest / need |
|---|---|---|
| General Manager / Executive | Decision maker | Overall performance KPIs and trends across locations |
| Revenue / Pricing Manager | Primary user of model | Price drivers and price predictions to set competitive rates |
| Marketing / Destination Manager | Planner | Which locations, segments, and seasons perform best |
| Property Managers | Daily user | Performance and guest ratings by property and location |
| Hosts / Property owners | Indirect user | Guidance on realistic pricing |
| Guests / Travelers | Indirect beneficiary | Better matched, fairly priced offers |
| Data / IT team | Maintainer | Reproducible pipeline, deployed service, documentation |
| DEPI Trainer / Evaluation Committee | Evaluator | Deliverables, quality, individual contribution |

## 1.6 Data Sources

| # | Source | Type | What it provides | Access / license |
|---|---|---|---|---|
| 1 | Inside Airbnb – https://insideairbnb.com/get-the-data/ | CSV into SQL | Listings, calendar availability, reviews for Barcelona, Amsterdam, Cape Town, Athens | Public download; follow site license terms |
| 2 | World Bank WDI – https://databank.worldbank.org/source/world-development-indicators | REST API / file | GDP per capita, tourism arrivals, consumer price inflation | Public; World Bank terms |
| 3 | Wikipedia | Web scraping | Landmarks and coordinates; distance-to-nearest-landmark features | Public; follow terms and credit source |
| 4 | Open-Meteo Historical Weather API – https://archive-api.open-meteo.com | REST API | Daily historical weather by latitude/longitude and date | Public API; follow terms |

Selected cities: Barcelona (Spain), Amsterdam (Netherlands), Cape Town (South Africa), and Athens (Greece). Backup cities: Rome (Italy) and Lisbon (Portugal).

## 1.7 Methodology

CRISP-DM: Business Understanding → Data Understanding/Collection → Data Preparation → Modeling → Evaluation → Deployment/Communication. Work is tracked in GitHub Projects. Code is managed with feature branches and pull requests.

## 1.8 Expected Deliverables

A professional portfolio repository containing: Business Case, Data Sources, SQL, Data Collection, Data Cleaning, EDA, Visualization, Machine Learning, Model Evaluation, Dashboard, Deployment, Documentation, and Stakeholder Presentation.

## 1.9 Assumptions & Constraints

- The team has six members.
- Only publicly available data sources are used.
- Inside Airbnb prices are asking prices, not confirmed transaction prices. Demand is represented by indirect indicators (availability and review counts), not actual bookings.
- World Bank indicators are country-level; with four countries, each indicator has few distinct values and mainly provides economic context and a coarse model feature.
- Deployment will use free tiers or student credits where possible. Cloud provider/account access is undecided.
- Final project discussion date is unknown and will be added when the training schedule is confirmed.

# 2. Project Plan

## 2.1 Milestones

Dates follow the DEPI schedule in the Documentation Guidelines.

| Milestone | Description | Due date |
|---|---|---|
| M1 | Planning & Management, Literature Review, Requirements Gathering submitted on GitHub | 16 Oct 2026 |
| M2 | System Analysis & Design submitted | 6 Nov 2026 |
| M3 | Implementation (source code & execution, including deployment) complete | 30 Nov 2026 |
| M4 | Final Presentation, Testing & Reports submitted | 4 Dec 2026 |
| M5 | Final project discussion | Unknown – pending training schedule |

## 2.2 Project Timeline

- Documentation and technical work intentionally overlap.
- Setup: choose cities and confirm data sources; set up repository and roles; notify trainer.
- Documentation: planning, literature review, requirements; system analysis and design; testing/QA documentation; user manual, presentation, infographic.
- Technical work: Phase 2 Data Collection; Phase 3 Data Preparation; Phase 4 EDA; Phase 5 Visualization & Dashboard; Phase 6 Machine Learning with MLflow; Phase 7 Evaluation; Phase 8 Deployment (API, Docker, CI, cloud).
- Timeline markers in the original plan: 11 Oct, 18 Oct, 25 Oct, 1 Nov, 8 Nov, 15 Nov, 22 Nov, and 29 Nov; M1–M4 dates are listed above.

## 2.3 Deliverables per Phase

| Phase | Deliverable | Owner | Due |
|---|---|---|---|
| Phase 1 | Business case (problem, KPIs, questions, success criteria) | Mariam – Team Lead | 16 Oct 2026 |
| Phase 2 | SQL scripts + collection notebooks + data sources documentation | Asmaa – Data Engineer | 24 Oct 2026 |
| Phase 3 | Cleaning notebook + processed dataset | Ahmed Nassar – Data Preparation, QA CI | &3 Nov 2026 |
| Phase 4 | EDA notebook + findings summary | Mostafa – Research Analyst | 10 Nov 2026 |
| Phase 5 | Dashboard + screenshots | Rofida – Visualization & UX | 20 Nov 2026 |
| Phases 6–7 | ML notebook + MLflow runs + evaluation report | Ahmed – ML Engineer | 27 Nov 2026 |
| Phase 8 | FastAPI app, Dockerfile, GitHub Actions workflow, cloud link | Ahmed , Ahmed, Mariam | 30 Nov 2026 |
| Phase 9 | Executive summary, technical report, presentation | Mariam (coordination); all members contribute | 4 Dec 2026 |

## 2.4 Resource Allocation

| Resource | Details / status |
|---|---|
| People | 6 team members |
| Languages / libraries | Python, SQL, pandas, scikit-learn |
| Database | PostgreSQL |
| Dashboard tool | Power BI |
| MLOps / Deployment | MLflow, FastAPI, Docker, GitHub Actions; cloud provider and account/credits undecided |
| Collaboration | GitHub, WhatsApp, GitHub Projects |
| Data | Inside Airbnb, World Bank WDI, Wikipedia, Open-Meteo |
| Compute | Local machines |

## 2.5 Team Workflow

- Weekly meeting: Friday at 9:00 PM, 30–45 minutes, to review progress, blockers, and next week's tasks.
- Branching: use feature branches (feature/<name>); no direct pushes to main.
- Pull requests: each PR is reviewed by at least one other member before merging.
- Commit messages: short and meaningful (e.g., “add: median imputation for price”).

• Trainer reporting: weekly deliverables are submitted as agreed with the trainer.

# 3. Task Assignment & Roles

Every member has a technical responsibility because each member's contribution is presented and evaluated individually. Backup assignments have not been provided and are marked as not assigned.

| # | Member | Role | Technical responsibility | Documentation responsibility | Backup |
|---|---|---|---|---|---|
| 1 | Mariam | Team Lead & Business Analyst | Phase 1; repository management; PR review; cloud account setup | Proposal, Project Plan, Roles, Risks, KPIs, README; final presentation integration | Not assigned |
| 2 | Mostafa | Research Analyst | Requirements; Phase 4 – EDA | Stakeholder analysis, user stories, requirements, Literature Review | Not assigned |
| 3 | Ahmed Nassar | Data Preparation, QA & CI | Phase 3 – integration, cleaning, normalization, encoding; Docker + GitHub Actions + automated tests | Test plan, test cases, bug reports | Not assigned |
| 4 | Ahmed | ML Engineer | Phases 6–7; MLflow + FastAPI | ML technical report, model evaluation, API documentation | Not assigned |
| 5 | Rofida | Visualization & UX | Phase 5 – dashboard and its hosting/link | Wireframes, UI/UX guidelines, user manual, infographic | Not assigned |
| 6 | Asmaa | Data Engineer (SQL & Collection) | Phase 2 – SQL database, APIs, Wikipedia scraping | Data sources, ERD, schema, data dictionary | Not assigned |

Shared responsibility: architecture, deployment diagram, DFD, and sequence/activity diagrams (Mariam, Asmaa, and Ahmed). Each member prepares and presents their own part in the final presentation.

# 4. Risk Assessment & Mitigation Plan

Probability and impact are rated Low / Medium / High.

| ID | Risk | Prob. | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| R1 | Late start / lost time before first deadline | High | High | Parallel documentation and technical work; daily check-ins until 16 Oct | Mariam |
| R2 | Data source unavailable, rate-limited, or changes format | Medium | High | Save raw snapshots; identify a backup source | Asmaa |
| R3 | Wikipedia scraping blocked, or landmark pages differ by city | Medium | Medium | Scrape politely with delays; check terms; use the official Wikipedia API or a manually verified landmark list per city | Asmaa |
| R4 | Data quality issues (missing values, inconsistent formats, outlier prices) | High | Medium | Document cleaning decisions; compare imputation/outlier options | Ahmed Nassar |
| R5 | Sources cannot be joined due to missing common keys | Medium | High | Verify join keys (city, date, latitude/longitude) before building the pipeline | Asmaa |
| R6 | Models perform poorly | Medium | Medium | Try several models and feature sets; report honestly with limitations | Ahmed |
| R7 | Member unavailable or not contributing | Medium | High | Assign backups; check weekly progress; inform trainer early | Mariam |
| R8 | Merge conflicts / lost work | Medium | Medium | Feature branches, small PRs, frequent commits | Mariam |
| R9 | Secrets (API keys, cloud credentials) pushed to GitHub | Medium | High | Use .gitignore, environment variables, and GitHub Secrets | Everyone |
| R10 | Deployment consumes time needed for core work | High | High | Start Phase 8 only after a working model exists; keep it minimal (one prediction endpoint, one container) | Mariam |
| R11 | Cloud account, credits, or free-tier limits block deployment | Medium | High | Confirm access early; keep Docker-based local deployment as fallback; document both | Mariam |
| R12 | Deployed service down or slow on presentation day | Medium | High | Test one day before; keep a recorded demo and local run as backup | Ahmed |
| R13 | World Bank country-level indicators add little signal | High | Medium | Use year-varying values where possible and report their contribution honestly | Asmaa |
| R14 | Listed prices differ from real transaction prices; demand not directly observable | High | Medium | Document as a limitation; use availability and review counts only as indirect indicators | Mostafa |
| R15 | Large Inside Airbnb files slow processing or exceed GitHub limits | Medium | Medium | Limit cities; keep large raw files out of the repository (.gitignore); document download steps | Asmaa |

# 5. Project KPIs

These measure project success. Business KPIs (e.g., average price and rating trends) belong in the Business Case. Targets are proposed by the team and may be revised.

| Category | KPI | How to measure | Target |
|---|---|---|---|
| Delivery | On-time submission of milestones | Milestones submitted by due date / total milestones | 100% |
| Delivery | Task completion | Completed tasks / planned tasks (weekly) | ≥ 90% |
| Collaboration | Contribution balance | Every member has meaningful commits and PRs | All 6 members |
| Data | Source integration | Sources successfully integrated | 4 of 4 sources |
| Data | Data completeness after cleaning | % non-missing in key columns | ≥ 95% |
| Modeling | Models compared | Number of models trained and evaluated | ≥ 3 |
| Modeling | Model performance | $R^2$, RMSE, MAE vs. a simple baseline | Better than baseline |
| Modeling | Experiment tracking | Share of model runs logged in MLflow | 100% |
| Dashboard | Required views present | KPIs, trends, comparisons, map, filters | All present |
| Dashboard | Load time | Seconds to load main view | ≤ 5 s |
| Deployment | Prediction API response time | Average seconds per request | ≤ 1 s |
| Deployment | Service availability | Reachable during demo/evaluation period | Reachable |
| Deployment | CI pipeline health | GitHub Actions checks pass on merges to main | Passing |
| Reproducibility | Fresh-install run | A teammate runs the project from README (including Docker) | Succeeds |
| Documentation | Coverage | Required documents completed / required documents | 100% |

# Open Items / To Confirm

- Cloud hosting provider and whether a student/free-tier account or credits are available.
- Final project discussion date, once the DEPI training schedule is confirmed.
- Backup person for each team member, if backups are to be assigned.